



RECONNAITY

AI-POWERED RECONNAISSANCE BROWSER EXTENSION

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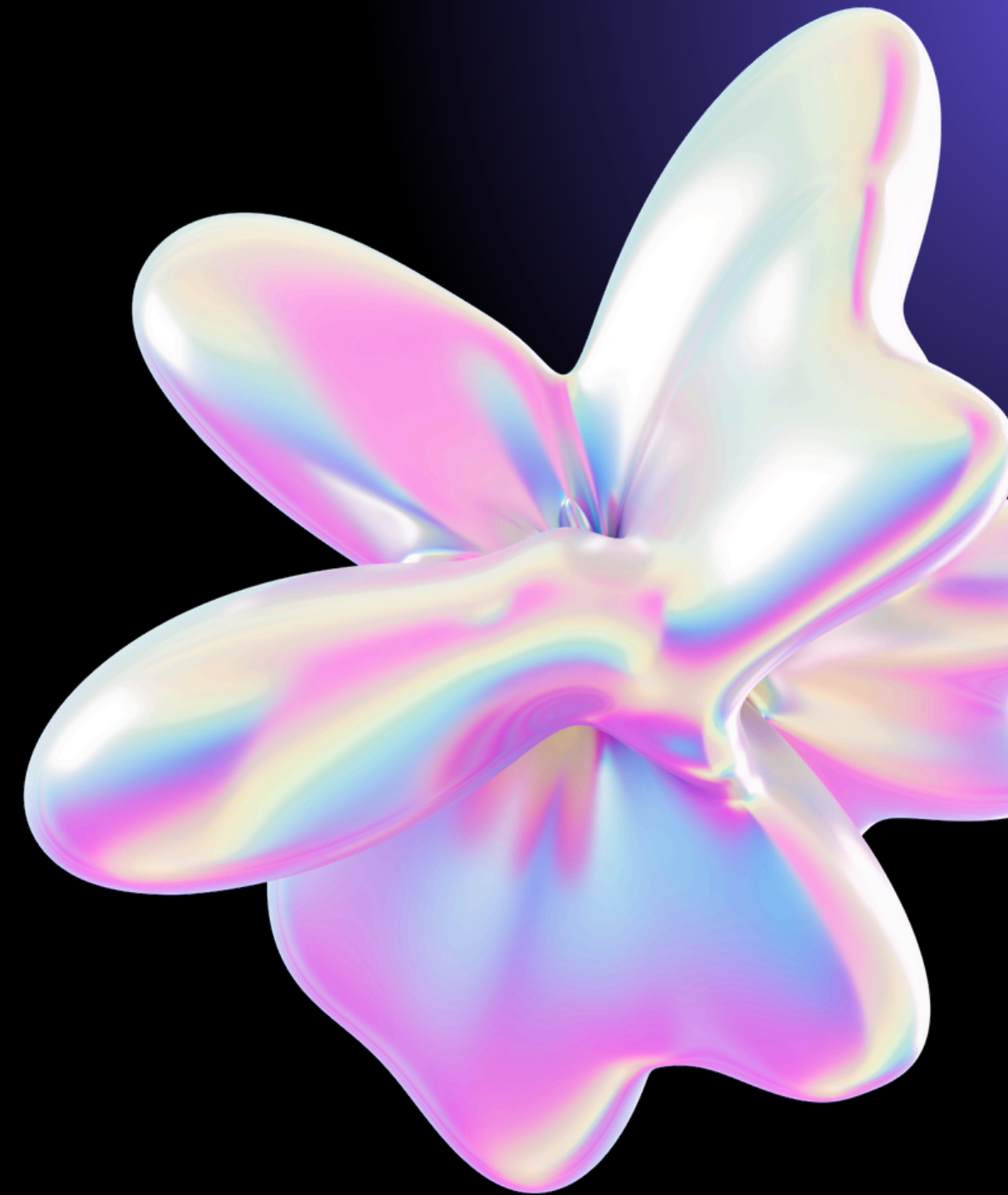
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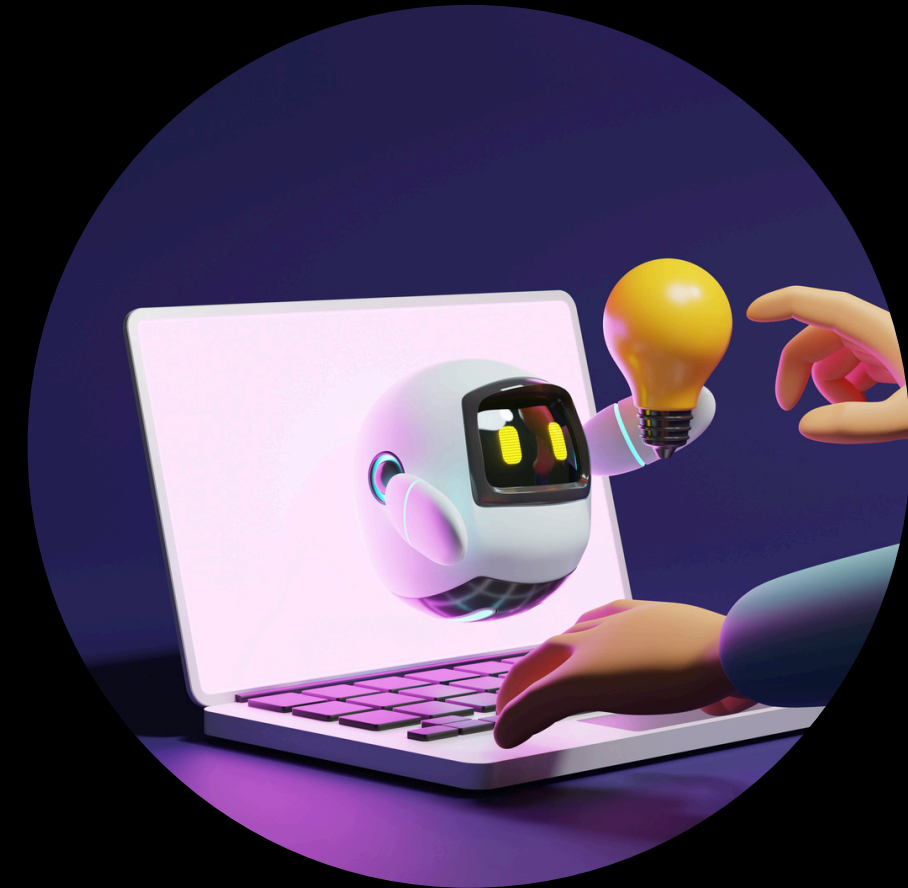
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Introduction

General problem area



Alarming Statistics

- 60% of breaches exploit unpatched software vulnerabilities.
- 90% of apps have critical security misconfigurations.

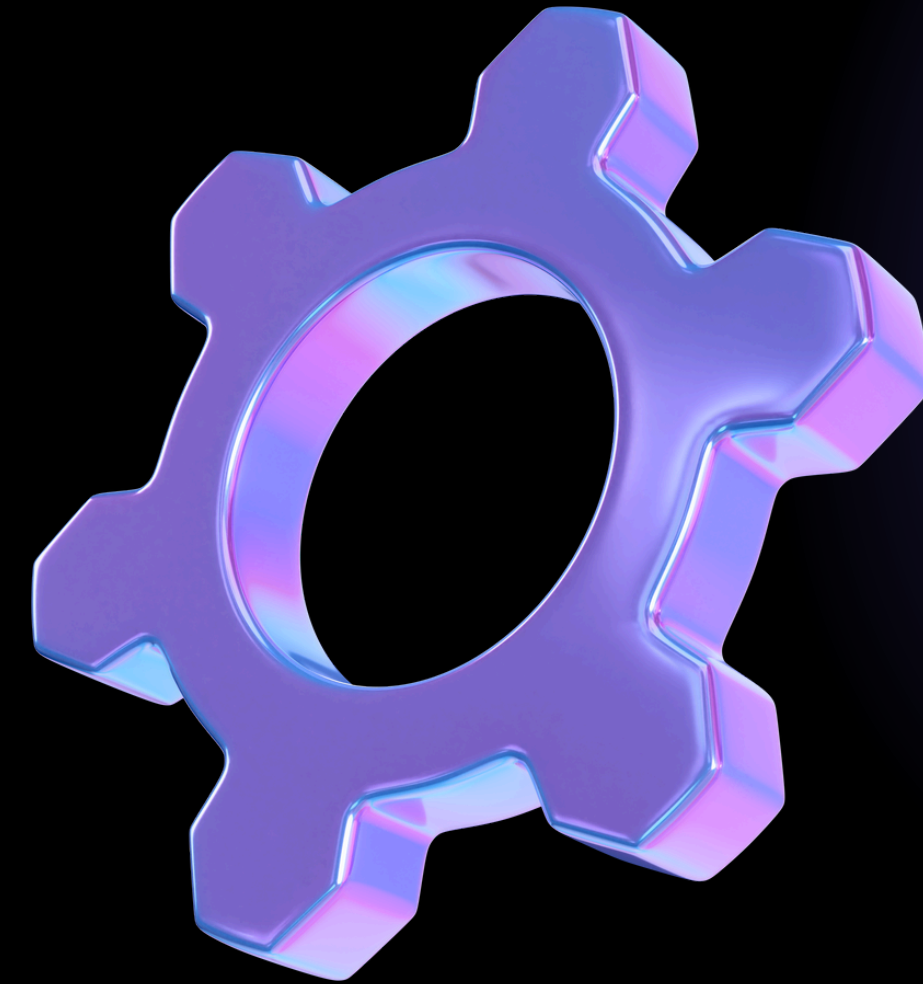
Websites - attacker favorite

- 50% of websites had serious exploitable vulnerabilities in 2021.
- 6 out of 18 components per site were outdated.
- Websites are prime targets due to public accessibility.

Consequences

- Financial Loss: Average breach cost \$4.8 million USD.
- Reputational Damage: 58% of customers lose trust after a breach.

Specific Problem addressing



Critical Gap

- Fragmented Tools: Existing solutions lack integration, requiring manual correlation of data.
- Actionable Insights Deficit: Most tools output raw data without contextualized insights .
- Offense-Defense Imbalance: Tools focus solely on offensive or defensive use cases, lacking unified workflows.

Project Goals & Problem Statement

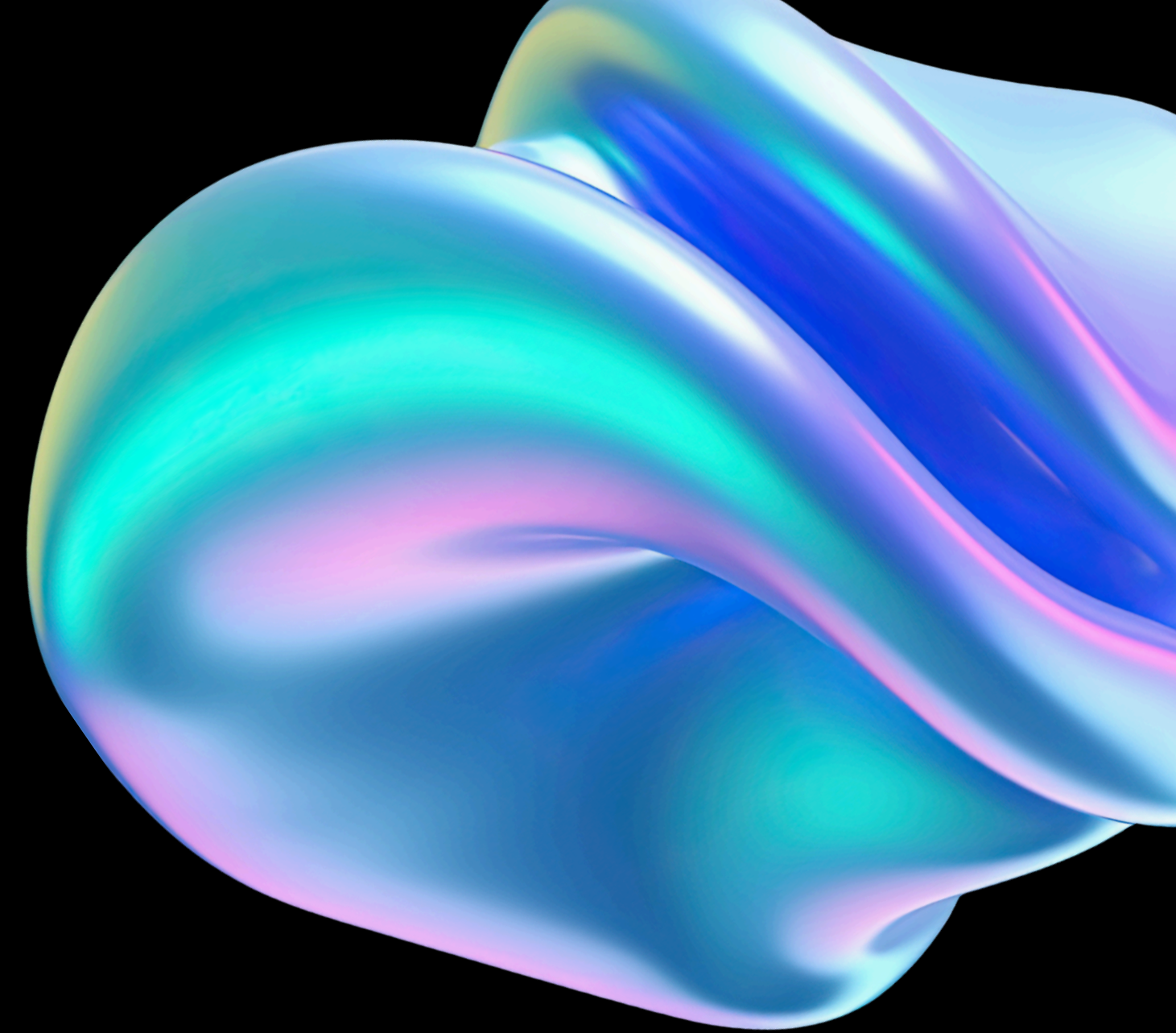
Websites remain highly vulnerable due to outdated, exploitable components. Ethical hacking's reconnaissance phase—critical for proactive risk identification—is hampered by tools that lack integration, create noise, and offer no actionable insights, leaving vulnerabilities undetected, unprioritized, and exploitable by attackers.

**Automate Stealthy
Comprehensive
Reconnaissance**

**Identify Risks
Efficiently**

**Integrate AI for
Actionable Insights**

**Bridge Offense-
Defense Workflows**



Existing Solutions

Vulnerability Scanners (e.g., Nessus, OpenVAS)

- Strengths: Broad CVE detection, CVSS scoring.
- Limitations: Heavy reliance on active scanning

Passive Recon Tools (e.g., Recon-ng)

- Strengths: Modular & broad: Passive data collection (WHOIS, DNS, IP) via modules.
- Limitations: CLI-dependent, no CVE links and outputs raw, unprioritized data.

Tech Stack Detectors (e.g., Wappalyzer)

- Strengths: Lightweight detection of tech stack.
- Limitations: No CVE correlation or actionable security guidance.

AI-Driven Tools (e.g., IBM QRadar)

- Strengths: Predictive threat modeling.
- Limitations: Enterprise-focused, cost-prohibitive, and not domain-specific.

Proposed Solution

RECONNAITY is a stealth-first, AI-powered browser extension. It integrates subdomain enumeration, technology detection, CVE detection and other data into a single click approach. Leveraging AI, it generates actionable insights for both technical teams and executives. Insuring automating complex workflows and enabling low-noise, real-time recon.

ALMOST ZERO-NOISE DATA COLLECTION

Leverages certificate transparency logs, public records, and passive techniques — no direct interaction with targets.

AI-POWERED ANALYSIS

Analyzes all collected data to generate offensive, defensive, and non-technical insights .

Proposed Solution

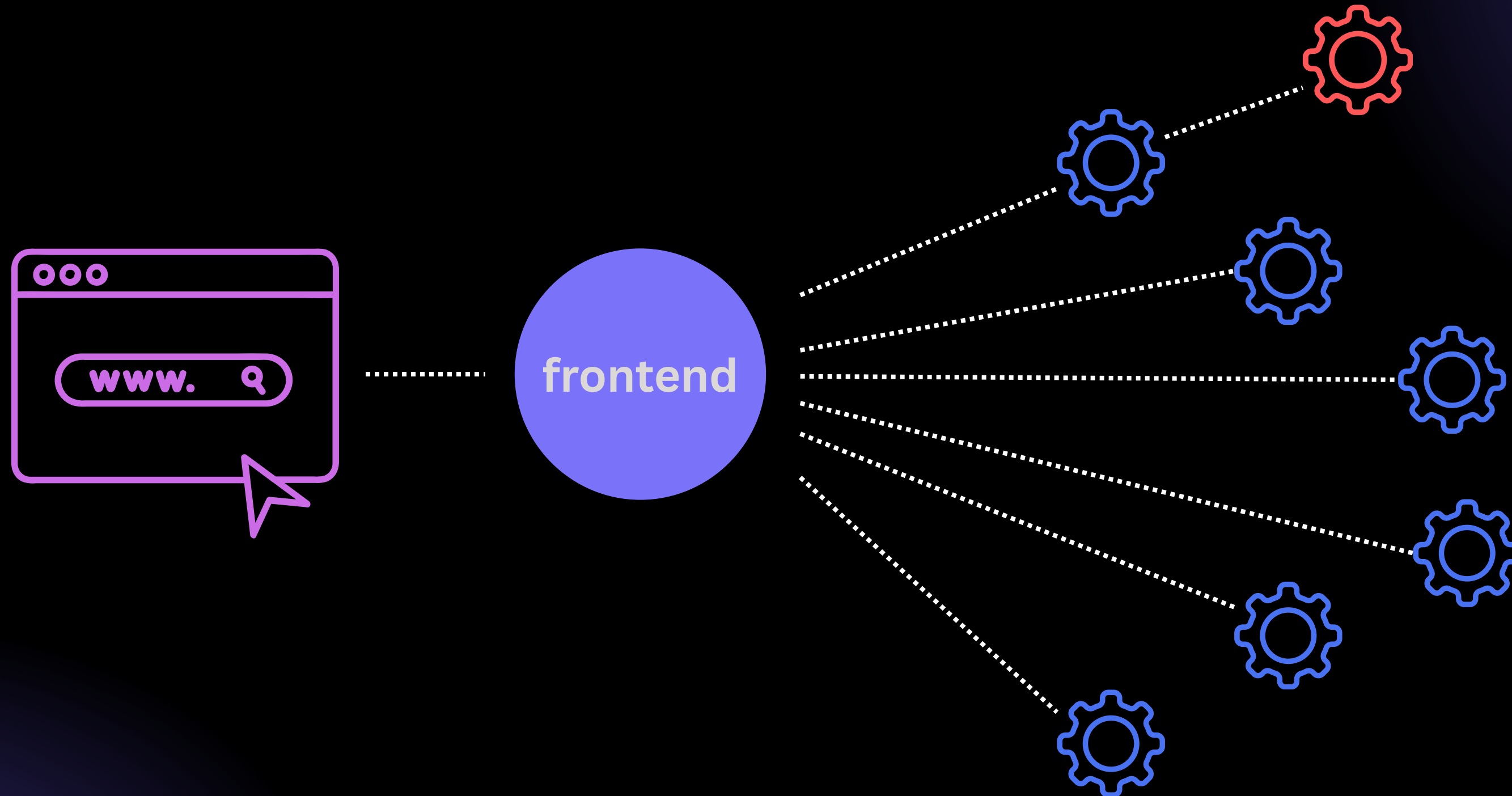
CUSTOM VULNERABILITY DETECTION

Use a proprietary algorithm to mapping detected technologies with a local NVD CVE database for version-specific vulnerability identification.

UNIFIED WORKFLOW

- One-Click Analysis.
- Dual Reporting.

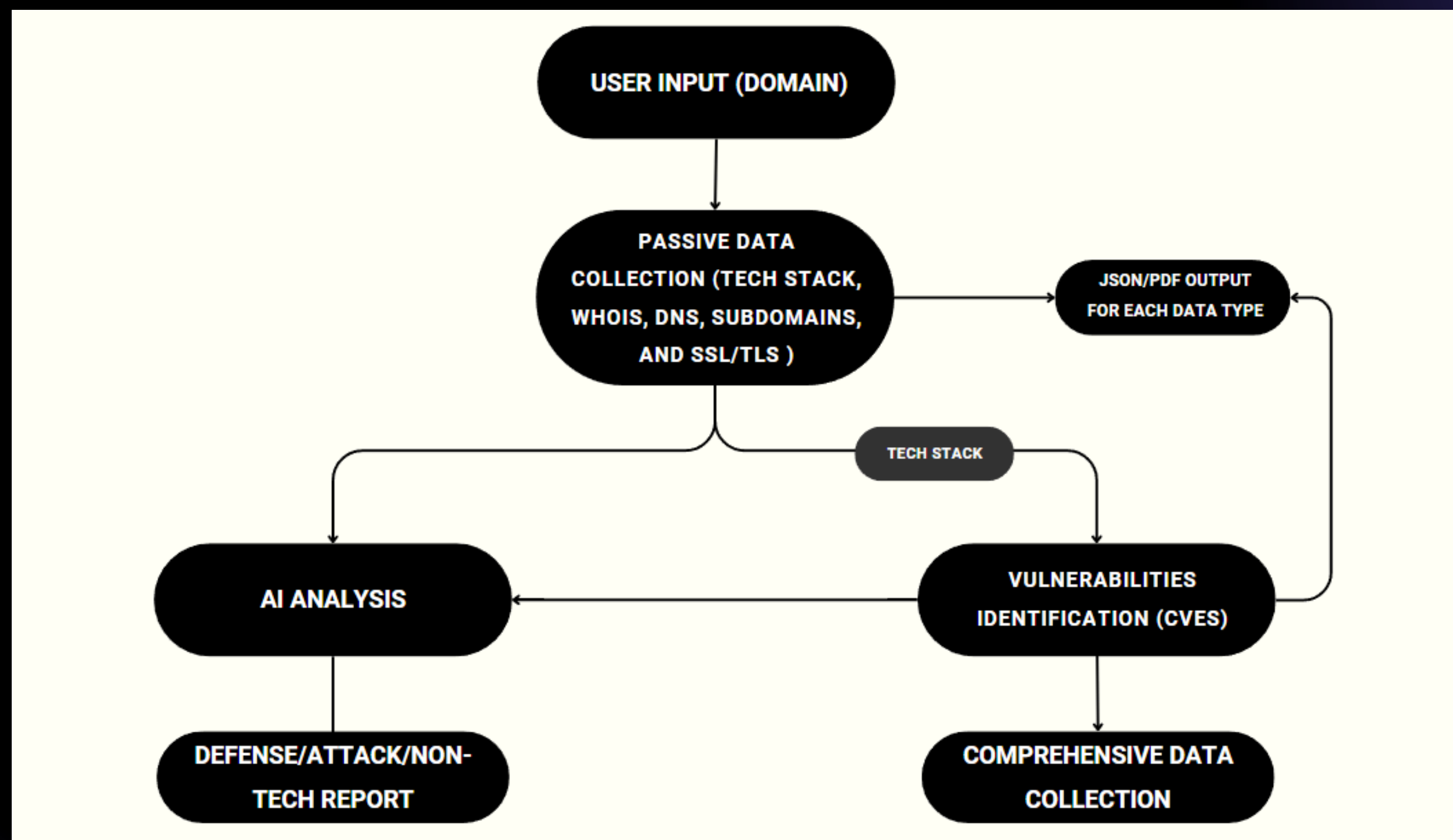
Architecture



Workflow Phases

PASSIVE DATA
COLLECTION +
VULNERABILITIES
IDENTIFICATION:
30 - 35 SECONDS
MAX

AI ANALYSIS:
80 - 40
SECONDS



Methodology Breakdown

BROWSER EXTENSION IMPLEMENTATION

- **Frontend:** Chromium extension with minimal UI for domain input and report download.
- **Backend:** Python APIs (FastAPI) handling data collection, analysis, and AI integration.

INPUT HANDLING

- **User Input:**
 - Target domain (e.g., example.com) is submitted via a Chromium browser extension (Manifest V3).
 - **Validation:**
 - Sanitizes input to block command injection.
 - Check if domain is reachable.

STEALTHY DATA COLLECTIONING

- **Subdomains**
 - **Tools:** Certificate Transparency (CT) Logs, Sublist3r.
 - **Method:** Enumerates subdomains from public CT logs and Sublist3r (tweaked to become passive).
- **DNS Records**
 - **Tools:** Python dns.resolver.
 - **Method:** Queries A, MX, TXT, CNAME records.
- **WHOIS Data**
 - **Tools:** Python whois library.
 - **Method:** Retrieves domain registration details.
- **SSL/TLS**
 - **Tools:** SSL Labs API, CT logs, custom algorithm
 - **Method:** Validates certificates and checks encryption protocols.

- **Tech Stack Detection:**
 - **Methods: Wappalyzer and HTTP headers identify technologies and their exact versions.**

VULNERABILITIES IDENTIFICATION

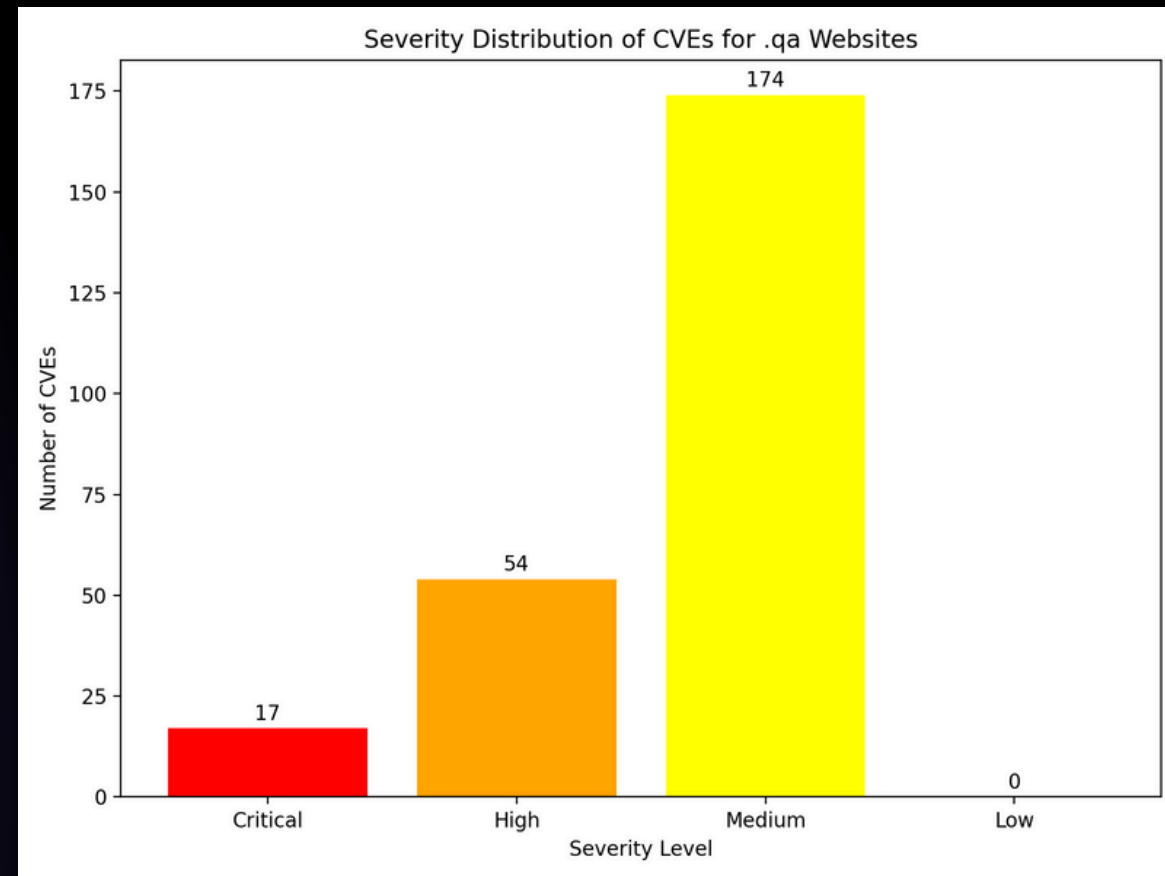
- **Custom Algorithm for CVE Matching:**
 - **Queries a locally stored NVD CVE database (regularly updated) to find vulnerabilities specific to detected software versions.**

AI-DRIVEN INSIGHTS

- **Model:** deepseek-r1-distill-llama-70b.
- **Input:** Structured data (CVEs, tech stacks, DNS anomalies, etc) + Our Custom-Developed AI Prompt.
- **Output:**
 - **Attack Roadmap.**
 - **Defense Roadmap.**
 - **Non-Technical Summary.**

OUTPUT GENERATION

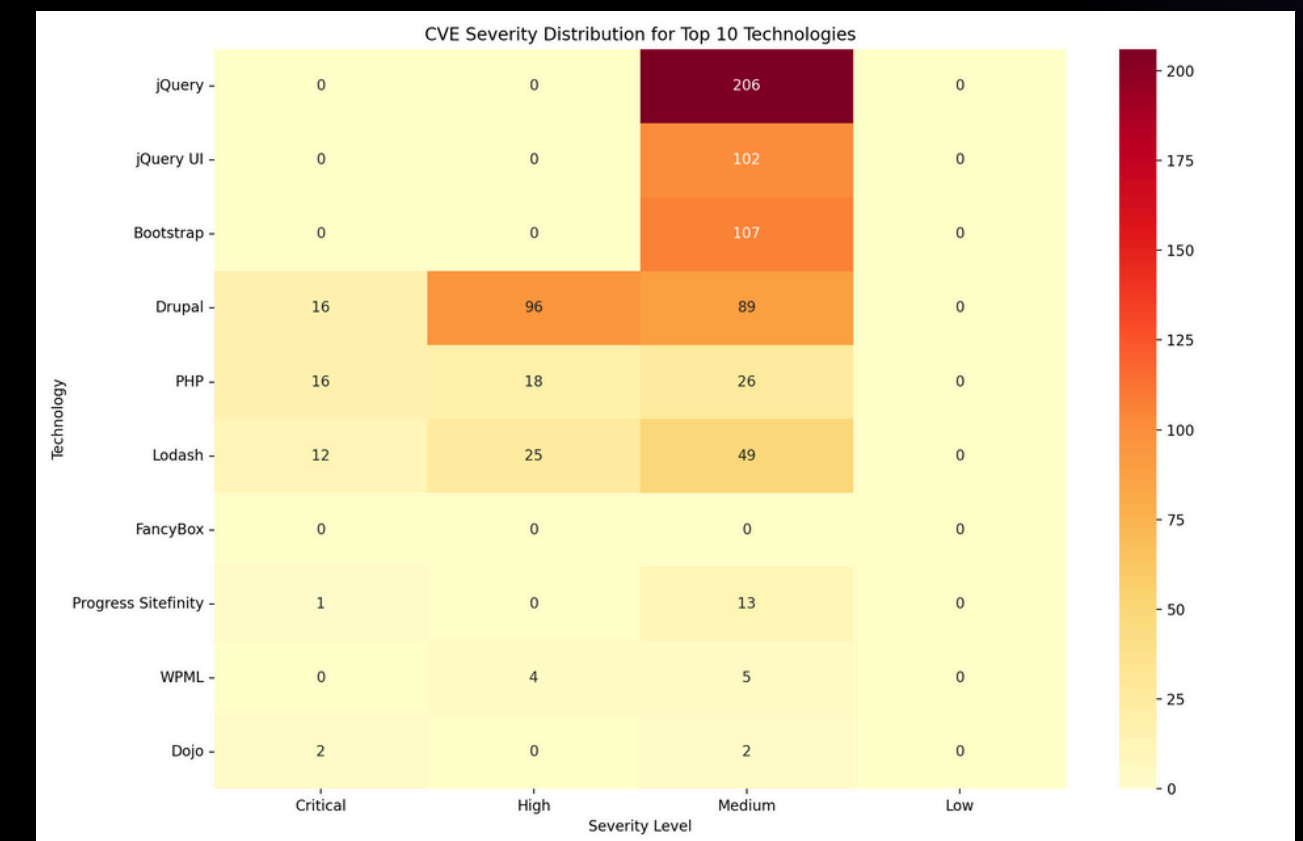
- **Technical Report (JSON):**
 - **Structure:** Machine-readable format for SIEM tools or pentesting pipelines.
- **User-Friendly Report (PDF):**
 - **Tools:** WeasyPrint + Markdown2.

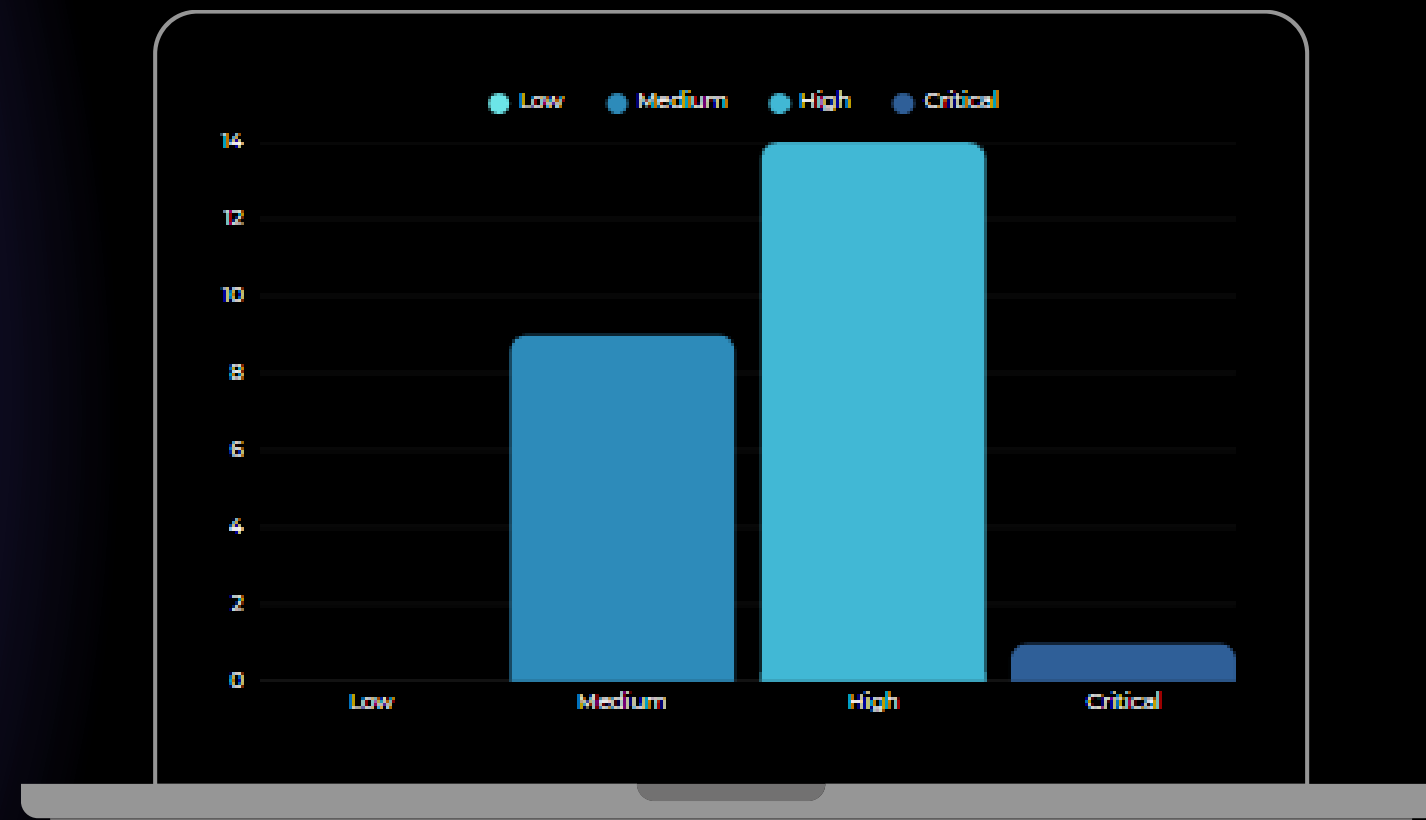


Testing & Finding

We audited 300+ random websites and 68 .qa websites to assess vulnerabilities.

- 01 42.6% of .qa websites have at least one CVE.
- 02 High/Critical severity vulnerabilities creates a big portion, with 54 High and 17 Critical CVEs identified.
- 03 Through nearly 300+ total websites we found technologies like Drupal and PHP contribute to 112 Critical/High CVEs.





UDST website testing

The website <http://udst.edu.qa/> stands out as one of the top 5 most vulnerable .qa websites we tested, with 24 total CVEs identified. Key findings include:

Severity Breakdown:

- 1 Critical
- 14 High
- 9 Medium
- 0 Low

The high concentration of Critical/High-severity vulnerabilities (15 total) suggests urgent patching is required.

The image shows a web browser displaying the University of Doha website (udst.edu.qa). The website features a header with navigation links: "Current Student", "Professional Education", "Alumni", "ADMISSIONS", "ACADEMIC", "LIFE AT UDST", and "RES". The main content area includes the university's logo and name in Arabic and English, along with a large banner image of a medical professional attending to a patient. Overlaid on the right side of the browser window is a white "RECONNAITY" interface. This interface includes a settings icon, a domain name input field, and three dropdown menus for "Mode" (set to "Simple"), "Process" (set to "Select process"), and "Format" (set to "Select format"). A blue "Analyze" button is located at the bottom of the overlay.

udst.edu.qa

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Current Student Professional Education Alumni

ADMISSIONS ACADEMIC LIFE AT UDST RES

Welcome to University of
for Science and Technol

DISCOVER MORE

RECONNAITY

Enter domain name...

Mode
Simple

Process
Select process

Format
Select format

Analyze

Conclusion

RECCONAITTY helps security teams quietly check websites for hidden risks, like outdated software or weak spots hackers could exploit. It provides clear steps to fix problems before they're abused. Think of it as a "digital detective" that makes the internet safer, faster, and cheaper to protect. For everyone else, it means fewer data breaches and more secure online experiences.

References

- GitHub. (Apr 19, 2023). tomnomnom/wappalyzer, GitHub. from <https://github.com/tomnomnom/wappalyzer>
- OWASP. (2021). OWASP Top 10. OWASP, from <https://owasp.org/Top10/>
- MITRE. (2025). Enterprise ATT&CK Matrix. MITRE ATT&CK, from <https://attack.mitre.org/matrices/enterprise/>
- PyPI. (Apr 10, 2025). Wappalyzer. Python Package Index, from <https://pypi.org/project/wappalyzer/>
- Whois. (n.d.). WHOIS lookup. WHOIS, from <https://www.whois.com/>
- GitHub. (Jul 29, 2020). aboul3la/Sublist3r. GitHub, from <https://github.com/aboul3la/Sublist3r>
- OpenRouter. (n.d.). AI models directory. OpenRouter, from <https://openrouter.ai/models>
- SSL Labs. (n.d.). SSL Labs APIs. Qualys, from <https://www.ssllabs.com/projects/ssllabs-apis/>
- crt.sh. (2025). Certificate Transparency Logs. crt.sh, from <https://crt.sh/>
- Bitdefender. (2019, December 1). 60% of breaches in 2019 involved unpatched vulnerabilities. Bitdefender Business Insights. Retrieved from <https://www.bitdefender.com/en-us/blog/businessinsights/60-of-breaches-in-2019-involved-unpatched-vulnerabilities>
- SentinelOne. (November 19, 2024). Security misconfiguration. SentinelOne Cybersecurity 101, from <https://www.sentinelone.com/cybersecurity-101/cybersecurity/security-misconfiguration/>
- Tripwire. (May 12, 2019). Unpatched vulnerabilities and breaches. State of Security, from <https://www.tripwire.com/state-of-security/unpatched-vulnerabilities-breaches>
- Help Net Security. (2024, January 25). Cybersecurity attack attempts increase. Help Net Security. Retrieved from <https://www.helpnetsecurity.com/2024/01/25/cybersecurity-attack-attempts-increase>
- Security Boulevard. (2024, December 1). Impact of unpatched vulnerabilities in 2025. Security Boulevard. Retrieved from <https://securityboulevard.com/2024/12/impact-of-unpatched-vulnerabilities-in-2025/>
- BankInfoSecurity. (August 16, 2024). Unpatched vulnerabilities cause 60% of cyber compromises. BankInfoSecurity, from <https://www.bankinfosecurity.com/unpatched-vulnerabilities-cause-60-cyber-compromises-a-26051>
- PDQ. (March 27, 2024). Risks of unpatched software vulnerabilities. PDQ Blog, from <https://www.pdq.com/blog/risks-of-unpatched-software-vulnerabilities/>
- CrowdStrike. (n.d.). Global threat report. CrowdStrike, from <https://www.crowdstrike.com/en-us/global-threat-report/>
- OWASP. (2021). A06:2021—Vulnerable and outdated components. OWASP Top 10, from https://owasp.org/Top10/A06_2021-Vulnerable_and_Outdated_Components/
- OWASP. (2021). A05:2021—Security misconfiguration. OWASP Top 10, from https://owasp.org/Top10/A05_2021-Security_Misconfiguration/
- Vercara. (2024, December 16). New Vercara research reveals impact of trust in brands following breaches, concerns around outside threats. Vercara. from <https://vercara.digicert.com/news/new-vercara-research-reveals-impact-of-trust-in-brands-following-breaches-concerns-around-outside-threats>
- PurpleSec. (n.d.). 2024 cybersecurity statistics: The ultimate list of stats, data & trends. PurpleSec. from <https://purplesec.us/resources/cybersecurity-statistics/>

**Thank you for listening, If
you have any questions
please feel free to ask!**

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